

EX.NO: 1(a)	STUDY OF NETWORK COMMANDS
Date :	

AIM

To study and understand various **network commands** used for network configuration, connectivity testing, troubleshooting, and monitoring.

Objective

- To learn commonly used network commands.
- To identify the IP address and network configuration of a system.
- To test connectivity between computers.
- To trace the path to a remote host.
- To examine active network connections.
- To troubleshoot basic network problems.

Prerequisite Knowledge

- Computer networks
- IP addresses
- TCP/IP protocol
- LAN and WAN
- DNS
- Routers and switches
- Command Prompt/Terminal

Theory

Network commands are command-line utilities used to obtain information about a computer's network configuration and to diagnose network-related problems.

COMMANDS

arp -a

Description:

Displays the **Address Resolution Protocol (ARP) table**, showing the mapping between IP addresses and MAC addresses of devices known to the computer.

Use:

- Verifies IP-to-MAC address mappings.
- Helps troubleshoot local network communication.
- Can help identify unexpected devices on a local network.

hostname

Description:

Displays the name of the current computer.

Use:

Useful for identifying a particular computer in a network containing multiple hosts.

`ipconfig`

Description:

Displays the basic TCP/IP configuration of the system, including the IP address, subnet mask, and default gateway.

Use:

Commonly used to check the current network configuration.

`ipconfig /all`

Description:

Displays detailed TCP/IP configuration information, including the MAC address, DHCP status, DNS servers, adapter information, IP address, and gateway.

Use:

Used for detailed network analysis and troubleshooting IP configuration.

`ipconfig /renew`

Description:

Requests a new IP configuration from the DHCP server.

Use:

Useful when troubleshooting DHCP-related connectivity problems or obtaining updated network configuration.

`ipconfig /release`

Description:

Releases the current DHCP-assigned IP configuration.

Use:

Normally used before obtaining a new DHCP configuration or when disconnecting from a network.

`ipconfig /flushdns`

Description:

Clears the local DNS resolver cache.

Use:

Useful when troubleshooting problems caused by outdated or incorrect cached DNS information.

`nbtstat -a`

Description:

Displays the NetBIOS name table of a specified computer over TCP/IP.

Use:

Useful for troubleshooting NetBIOS name-resolution and connectivity problems in Windows networks.

netdiag

Description:

`netdiag` was a Windows network diagnostic utility used to perform various network diagnostic tests.

Use:

It was used for diagnosing network adapters, configurations, and connectivity.

netstat

Description:

Displays active network connections, listening ports, and network statistics.

For numerical addresses:

```
netstat -n
```

Use:

- Monitors active connections.
- Identifies listening ports.
- Helps troubleshoot TCP/UDP communication.

nslookup

Description:

Queries DNS servers to obtain information about domain names and IP addresses.

Command:

```
nslookup google.com
```

Use:

Essential for diagnosing DNS problems and verifying domain-name resolution.

pathping

Description:

Combines features of `ping` and `tracert`. It identifies the route to a destination and collects packet-loss and latency information.

Command:

```
pathping google.com
```

Use:

Helps identify network links or routers where packet loss or delays are occurring.

ping

Description:

Sends ICMP Echo Request messages to a destination and waits for Echo Replies.

Command:

```
ping google.com
```

Use:

- Tests network connectivity.
- Determines whether a host is reachable.
- Measures approximate round-trip response time.

route**Description:**

Displays and, when permitted, modifies the local IP routing table.

Command:

```
route print
```

Use:

- Examines routing information.
- Helps understand packet forwarding.
- Can be used to configure static routes.

tracert**Description:**

Traces the path taken by packets to reach a destination by displaying intermediate network hops.

Command:

```
tracert google.com
```

Use:

- Identifies intermediate routers.
- Helps locate routing problems.
- Helps identify network hops with high delay.

TCPDUMP COMMANDS

`tcpdump` is a command-line packet analyzer commonly available on **Linux/Unix-like systems**. It captures and analyzes network traffic.

a) Capture packets from all interfaces

```
tcpdump -i any
```

Use:

Captures packets from all available interfaces supported by the system.

b) Capture packets from a specific interface

```
tcpdump -i eth0
```

Use:

Captures traffic only from the specified network interface.

c) Read packets from a saved capture file

```
tcpdump -r packets_file
```

Use:

Allows previously captured packets to be analyzed offline.

d) Capture traffic from a network

```
tcpdump net 192.168.1.0/24
```

Use:

Captures traffic associated with the specified IP network.

e) Capture packets from a specific source

```
tcpdump src 192.168.1.100
```

Use:

Captures packets whose source matches the specified IP address.

f) Capture packets destined for a specific host

```
tcpdump dst 192.168.1.100
```

Use:

Captures packets whose destination matches the specified IP address.

g) Capture SSH traffic

```
tcpdump ssh
```

Use:

Filters traffic associated with SSH when the installed tcpdump supports the `ssh` protocol name.

h) Capture traffic on port 22

```
tcpdump port 22
```

Use:

Captures traffic using port 22, commonly associated with SSH.

i) Capture traffic within a port range

```
tcpdump portrange 22-125
```

Use:

Captures traffic using ports from 22 through 125.

S.No.	Command	Main Purpose
1	arp -a	Display ARP table
2	hostname	Display computer name
3	ipconfig	Display basic IP configuration
4	ipconfig /all	Display detailed IP configuration
5	ipconfig /renew	Renew DHCP configuration
6	ipconfig /release	Release DHCP configuration
7	ipconfig /flushdns	Clear DNS cache
8	nbtstat -a	Display NetBIOS information
9	netdiag	Network diagnostics on older Windows systems
10	netstat	Display network connections
11	nslookup	Perform DNS queries
12	pathping	Analyze route, delay, and packet loss
13	ping	Test connectivity
14	route print	Display routing table
15	tracert	Trace network path
16	tcpdump	Capture and analyze packets

Advantages

1. Easy to use from the command line.
2. Useful for network troubleshooting.
3. Provides network configuration information.
4. Helps identify connectivity problems.
5. Helps administrators monitor network connections.
6. Does not require a graphical interface.

Disadvantages

1. Some commands require administrator privileges.
2. Output can be difficult for beginners to understand.
3. Firewall restrictions may affect results.
4. Commands and options can differ between operating systems.

Real-Life Applications

- Troubleshoot Internet connectivity.
- Find IP and MAC addresses.
- Diagnose DNS problems.

- Check routing.
- Identify network delays.
- Monitor active connections.
- Locate network failures.

Viva Questions and Answers

1. What is a network command?

A network command is a command-line utility used to configure, monitor, or troubleshoot a network.

2. Which command displays the IP address?

`ipconfig`

3. Which command tests connectivity?

`ping`

4. Which command traces the route to a destination in Windows?

`tracert`

5. Which command is used for DNS lookup?

`nslookup`

6. Which command displays the MAC address?

`getmac`

7. What does ARP do?

ARP maps an IP address to a MAC address on a local network.

8. Which command displays active network connections?

`netstat`

9. Which command displays the routing table?

`route print`

10. What is the purpose of `pathping`?

It analyzes the path to a destination and provides information about packet loss and latency

Result

Thus, various network commands were studied and their functions, syntax, and applications in network configuration, connectivity testing, troubleshooting, routing, DNS analysis, and packet monitoring were understood.

EX.NO: 1(b)	SIMULATION OF PING COMMAND
Date:	

AIM

To write and execute a Java program to simulate the **PING command** and check whether a destination host is reachable.

Objective

- To understand the working of the PING command.
- To check network connectivity between two hosts.
- To determine the response time of a destination host.

Theory

PING (Packet Internet Groper) is a network utility used to test whether a host is reachable over a network.

It sends an **ICMP Echo Request** to the destination. If the destination is reachable, an **ICMP Echo Reply** is received.

In Java, the `InetAddress` class provides the `isReachable()` method, which can be used for a basic simulation of the ping operation.

Advantages

1. Simple and easy to implement.
2. Helps understand network connectivity.
3. Measures approximate response time.
4. Useful for basic network troubleshooting.

Disadvantages

1. `isReachable()` is not a complete replacement for the operating-system ping utility.
2. Results depend on the operating system and network configuration.
3. Firewalls may prevent responses.
4. It does not provide all ICMP packet details.

Applications

- Testing network connectivity.
- Checking whether a server is reachable.
- Troubleshooting network problems.
- Measuring approximate network response time.
- Testing local and remote hosts.

Algorithm

1. Start the program.

2. Read the destination host name or IP address.
3. Obtain the IP address using InetAddress.
4. Send a reachability request.
5. Record the start and end time.
6. Calculate the response time.
7. Display the result.
8. Repeat the process four times.
9. Stop the program.

JAVA PROGRAM (PING SIMULATION)

```
import java.net.InetAddress;

import java.util.Scanner;

public class PingSimulation {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter host name or IP address: ");

        String host = sc.nextLine();

        try {

            InetAddress address = InetAddress.getByName(host);

            System.out.println("\nPing " + host + " [" +

                address.getHostAddress() + "]");

            for (int i = 1; i <= 4; i++) {

                long start = System.currentTimeMillis();

                if (address.isReachable(5000)) {

                    long end = System.currentTimeMillis();

                    System.out.println("Reply from "

                        + address.getHostAddress()

                        + ": time=" + (end - start) + " ms");

                } else {
```

```
        System.out.println("Request timed out.");
    }

    Thread.sleep(1000);
}
} catch (Exception e) {
    System.out.println("Error: " + e.getMessage());
}
sc.close();
}
}
```

Run

```
Javac PingSimulation.java
java PingSimulation
```

Input

Enter host name or IP address: google.com

Output

Pinging google.com [64.233.181.113]

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Viva Questions and Answers

1. What is PING?

PING is a network utility used to check whether a host is reachable.

2. What does PING stand for?

Packet Internet Groper.

3. Which protocol is normally used by PING?

ICMP — Internet Control Message Protocol.

4. What is an Echo Request?

It is an ICMP message sent to check whether a destination is reachable.

5. What is an Echo Reply?

It is the response sent by the destination to an ICMP Echo Request.

6. Which Java class is used in this program?

`InetAddress`.

7. Which method is used to check reachability?

`isReachable()`.

8. What does `getHostAddress()` return?

It returns the IP address of the host.

9. What does "Request timed out" mean?

It means that a response was not received within the specified time.

10. What is response time?

It is the approximate time taken to receive a response from the destination.

Result

Thus, the Java program to simulate the PING command was implemented and executed successfully.

EX.NO: 1(c)	SIMULATION OF TRACEROUTE COMMAND
Date :	

AIM

To write a Java program to simulate the **Traceroute (tracert) command** and display the intermediate routers/hops between the source computer and a destination host.

Objective

- To understand the working of the traceroute command.
- To identify the intermediate network hops.
- To determine the IP address and response time of each hop.
- To understand how packets travel from source to destination.

Theory

Traceroute is a network diagnostic command used to determine the path followed by packets from a source computer to a destination.

In Windows, the command is called:

tracert

In Linux/macOS, it is commonly called:

traceroute

Traceroute works by sending packets with gradually increasing **TTL (Time To Live)** values. When the TTL becomes zero at an intermediate router, the router returns a response. By recording these responses, the path through the network can be identified.

Advantages

1. Easy to implement using Java.
2. Helps understand network routing.
3. Can identify whether a destination is reachable.
4. Measures approximate response time.
5. Useful for basic network troubleshooting.

Disadvantages

1. Java's `InetAddress.isReachable()` does **not provide a true hop-by-hop traceroute**.
2. It may not reveal every intermediate router.
3. Results depend on operating-system permissions and network configuration.
4. Firewalls may block ICMP or related responses.
5. Response time can vary depending on network traffic.

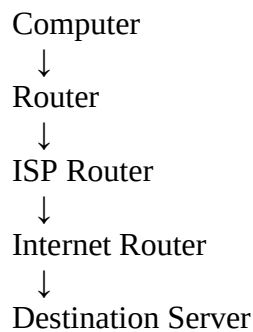
Applications

- Network troubleshooting.

- Finding routing problems.
- Measuring network response time.
- Studying computer networks.
- Identifying connectivity problems.

Real-Life Example

When a user cannot access a website, traceroute can help determine where the connection is failing:



By examining the hops, a network administrator can identify possible routing or connectivity problems.

ALGORITHM

1. Start the program.
2. Read the destination host name or IP address from the user.
3. Find the IP address of the destination.
4. Set the maximum number of hops.
5. For each hop:
 - Attempt to determine a reachable address.
 - Measure the response time.
 - Display the hop number, IP address, and response time.
6. Continue until the destination is reached or the maximum hop count is exceeded.
7. Stop the program.

JAVA PROGRAM (TRACEROUTE)

```
import java.net.InetAddress;

import java.util.Scanner;

public class TracerouteSimulation {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```

System.out.print("Enter destination host name or IP address: ");

String host = sc.nextLine();

try {

    InetAddress destination = InetAddress.getByName(host);

    System.out.println();

    System.out.println("Tracing route to " + host +
        " [" + destination.getHostAddress() + "]");

    System.out.println("Maximum number of hops: 10");

    System.out.println();

    for (int ttl = 1; ttl <= 10; ttl++) {

        long start = System.currentTimeMillis();

        boolean reachable = destination.isReachable(3000);

        long end = System.currentTimeMillis();

        System.out.print(ttl + "    ");

        if (reachable) {

            System.out.println(
                destination.getHostAddress() +
                "    " + (end - start) + " ms"
            );

            System.out.println();

            System.out.println("Trace complete.");

            break;

        } else {

            System.out.println("*    Request timed out.");

        }

    }

} catch (Exception e) {

```

```
        System.out.println("Error: " + e.getMessage());
    }

    sc.close();
}
}
```

Run

```
javac TracerouteSimulation.java
java TracerouteSimulation
```

Input

Enter destination host name or IP address: google.com

Output

Enter destination host name or IP address: google.com

Tracing route to google.com [142.251.184.101]

Maximum number of hops: 10

- 1 * Request timed out.
- 2 * Request timed out.
- 3 * Request timed out.
- 4 * Request timed out.
- 5 * Request timed out.
- 6 * Request timed out.
- 7 * Request timed out.
- 8 * Request timed out.
- 9 * Request timed out.
- 10 * Request timed out.

Viva Questions and Answers

1. What is traceroute?

Traceroute is a network diagnostic tool used to determine the path taken by packets to reach a destination.

2. What is the Windows command for traceroute?

`tracert`

3. What is TTL?

TTL stands for **Time To Live**. It limits the lifetime of a packet in a network.

4. Why is traceroute useful?

It helps identify intermediate routers and locate network connectivity problems.

5. Which Java class is used in this program?

`InetAddress` from the `java.net` package.

6. Which method checks whether a host is reachable?

`isReachable()`.

7. What does `getHostAddress()` return?

It returns the IP address of the host.

8. What does `getByName()` do?

It obtains an `InetAddress` object for a specified host name or IP address.

9. What does * indicate in traceroute?

It generally indicates that a response was not received within the specified timeout.

10. Is this program exactly the same as the operating-system traceroute command?

No. It is a **simulation/basic Java implementation**. A true traceroute requires controlling TTL values and processing ICMP/UDP responses for each individual hop.

Result

Thus, the Java program to simulate the Traceroute command was implemented and executed successfully.

EX. NO :2	CREATE SOCKET FOR HTTP FOR WEB PAGE UPLOAD AND DOWNLOAD
Date :	

AIM

To write a Java HTTP web client program that downloads a web page using TCP sockets.

Objectives

- Understand TCP socket communication.
- Establish a connection with a web server.
- Send an HTTP request.
- Receive an HTTP response.
- Display/download the webpage.

Prerequisite Knowledge

- TCP/IP.
- Client-server architecture.
- Socket programming.
- HTTP protocol.
- Java I/O streams.

Theory / Concept

HTTP operates as an application-layer protocol. A web client establishes a TCP connection with a web server and sends an HTTP request.

Typical request:

GET / HTTP/1.1

Host: example.com

Connection: close

The server responds with an HTTP response containing:

- Status line
- Headers
- Webpage content

Advantages

- Reliable communication through TCP.
- Simple client-server architecture.
- Useful for understanding HTTP internally.

Disadvantages

- Plain HTTP does not provide encryption.
- Manual HTTP parsing is more complex than using high-level libraries.
- Modern websites may require HTTPS.

Applications

- Web browsers.
- Web crawlers.
- Testing tools.
- HTTP learning applications.
- Network programming.

Real-Time Example

A browser connects to a web server and sends a GET request to retrieve the homepage.

ALGORITHM

1. Create a TCP socket.
2. Connect to the web server on port 80.
3. Create an HTTP GET request.
4. Send the request.
5. Read the server response.
6. Display the response.
7. Close the socket.

PROGRAM

```
import java.io.*;
import java.net.*;

public class HTTPClient {
    public static void main(String[] args) {
        String host = "example.com";
        try {
            Socket socket = new Socket(host, 80);
            PrintWriter out = new PrintWriter(
                socket.getOutputStream(), true);
            BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            out.println("GET / HTTP/1.1");
            out.println("Host: " + host);
```

```
        out.println("Connection: close");
        out.println();

        String response;
        while ((response = in.readLine()) != null) {
            System.out.println(response);
        }
        socket.close();
    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}
```

Run

```
javac HTTPClient.java
java HTTPClient
```

Input

Host: example.com
Port: 80
Request: GET /

Output

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: ...
<!doctype html>
<html>
...
</html>
```

Viva Questions

1. What is HTTP?

HTTP stands for HyperText Transfer Protocol.

2. Which transport protocol does HTTP normally use?

TCP for traditional HTTP/1.x and HTTP/2.

3. What is a socket?

A socket is an endpoint for network communication.

4. What is port 80?

The conventional port for HTTP.

5. What is GET?

An HTTP method used to request a resource.

6. What is a web server?

A system that serves web resources to clients.

7. What is an HTTP response?

The response returned by a server after receiving an HTTP request.

8. What is TCP?

A reliable, connection-oriented transport protocol.

9. What is localhost?

The local computer, commonly represented by 127.0.0.1.

10. What does Connection: close indicate?

The TCP connection should be closed after the response.

Result

Thus, an HTTP web client was successfully implemented using Java TCP sockets.